

RCIM Track1 Retrained Paper Tables Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Table Mapping

Paper table	Report equivalent
Table 2	Amplitude MAE by harmonic and family
Table 3	Amplitude RMSE by harmonic and family
Table 4	Phase MAE by harmonic and family
Table 5	Phase RMSE by harmonic and family

Sources

Dataset/input mode	Source evidence
Simplified Dataset + Setpoints	<code>models/simplified_dataset/paper_reference/rcim_track1</code>
Polished Dataset + Setpoints	<code>models/polished_dataset/paper_reference/rcim_track1/setpoints/promotion_inventory.yaml</code>
Polished Dataset + Actual Values	<code>models/polished_dataset/paper_reference/rcim_track1/actual_values/promotion_inventory.yaml</code>

Simplified Dataset + Setpoints

- Dataset: `simplified_dataset`
- Input mode: `setpoints`
- Metric rows loaded: `418`
- Source mode: `reference_inventory`

No `global` surface archive was found for this dataset/input mode; only `forward` and `backward` paper-table equivalents are reported.

Forward Surface

Family columns: SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM, ELM .

Table 2 Equivalent - Amplitude MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.00305677	0.0271161	0.0029211	0.00334252	0.00302836	0.0032981
ampl_1	1	6.56573e-05	0.0289634	3.17912e-05	3.19493e-05	3.40618e-05	2.99936e-05
ampl_3	3	7.61143e-05	0.013724	2.20406e-05	2.48493e-05	2.68639e-05	1.90726e-05
ampl_39	39	5.12792e-05	0.0150448	3.03707e-05	5.40545e-05	5.40808e-05	2.58271e-05
ampl_40	40	5.42676e-05	0.0229618	2.359e-05	3.3619e-05	3.97516e-05	2.06984e-05
ampl_78	78	0.000130489	0.0222392	5.99545e-05	9.67335e-05	0.000100007	3.40623e-05
ampl_81	81	4.18793e-05	0.0263144	1.40259e-05	1.78349e-05	2.35786e-05	1.13607e-05
ampl_156	156	0.000519983	0.0233848	0.000122525	0.000150126	0.000123802	5.42847e-05
ampl_162	162	0.000680543	0.0346719	0.000199132	0.000142924	0.000119119	8.02514e-05
ampl_240	240	0.000242616	0.0227904	8.14611e-05	8.49043e-05	0.000223229	3.96275e-05

Panel 2 of 2: GBM, HGBM, XGBM, LGBM, ELM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM	ELM
ampl_0	0	0.00342451	0.0032848	0.00409938	0.00608785	0.00636693
ampl_1	1	3.36664e-05	3.01984e-05	5.8509e-05	3.53012e-05	3.42378e-05
ampl_3	3	2.68844e-05	2.74001e-05	7.00635e-05	5.23761e-05	7.05152e-05
ampl_39	39	4.56841e-05	4.22901e-05	0.000112889	0.0001078	0.00010553
ampl_40	40	2.48912e-05	2.7153e-05	6.52937e-05	3.71074e-05	4.4305e-05
ampl_78	78	7.29048e-05	6.9168e-05	0.000130015	0.000176727	0.000253924
ampl_81	81	1.38016e-05	1.56039e-05	4.84409e-05	2.41069e-05	3.27129e-05
ampl_156	156	0.000169589	0.000192565	0.000213958	0.000334592	0.000661327
ampl_162	162	0.000225139	0.000331003	0.000250252	0.000453492	0.00095085
ampl_240	240	8.29394e-05	9.24011e-05	0.000172708	0.000153304	0.000313632

Table 3 Equivalent - Amplitude RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.00338731	0.0444009	0.0035838	0.00448775	0.00363044	0.00402504
ampl_1	1	8.19255e-05	0.0506314	4.03765e-05	4.32286e-05	4.51865e-05	3.95159e-05
ampl_3	3	8.95392e-05	0.0280048	3.01094e-05	3.36444e-05	3.77537e-05	2.5487e-05
ampl_39	39	6.45351e-05	0.0312168	3.91365e-05	6.90261e-05	7.3839e-05	3.32522e-05
ampl_40	40	7.34706e-05	0.035164	3.33647e-05	4.23337e-05	6.71039e-05	2.74003e-05
ampl_78	78	0.000159871	0.0532152	7.90751e-05	0.000121027	0.000137482	4.85815e-05
ampl_81	81	5.33825e-05	0.0407882	1.9156e-05	2.22354e-05	3.08913e-05	1.56353e-05
ampl_156	156	0.00110312	0.0512525	0.000372638	0.000387564	0.000365913	0.000226686
ampl_162	162	0.00225189	0.0782889	0.000615249	0.000556478	0.000307436	0.000326493
ampl_240	240	0.000607111	0.0353102	0.000155537	0.000119526	0.000607463	7.97047e-05

Panel 2 of 2: GBM, HGBM, XGBM, LGBM, ELM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM	ELM
ampl_0	0	0.00429076	0.00394434	0.00530128	0.00766252	0.00848451
ampl_1	1	4.40207e-05	3.92211e-05	7.53981e-05	4.64541e-05	4.29732e-05
ampl_3	3	3.51095e-05	3.57166e-05	9.14248e-05	6.46017e-05	9.23106e-05
ampl_39	39	5.66209e-05	5.29581e-05	0.000139317	0.000124724	0.000150356
ampl_40	40	3.373e-05	4.08969e-05	8.88706e-05	5.54452e-05	6.42287e-05
ampl_78	78	9.95071e-05	9.33812e-05	0.000177205	0.000210241	0.000341396
ampl_81	81	1.92187e-05	2.04346e-05	5.99168e-05	3.00738e-05	4.17627e-05
ampl_156	156	0.00036332	0.000417758	0.000416185	0.000543088	0.00104785
ampl_162	162	0.0006087	0.000840289	0.000653461	0.00117065	0.00228557
ampl_240	240	0.000184329	0.000190047	0.000308105	0.000291955	0.000596099

Table 4 Equivalent - Phase MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00293219	0.0183255	0.00191692	0.00249641	0.00241905	0.00213975
phase_3	3	0.0336004	0.0656061	0.0213333	0.0417107	0.0424304	0.0211093
phase_39	39	0.0378276	0.100654	0.0271451	0.0381584	0.0363294	0.0271126
phase_40	40	0.0992124	0.0875861	0.0354547	0.0568484	0.0682433	0.0327383
phase_78	78	0.163665	0.127547	0.070632	0.104668	0.0960632	0.0664968
phase_81	81	0.137422	0.175523	0.0583387	0.0985556	0.0900424	0.0456165
phase_156	156	2.005	1.7145	0.651807	0.83352	1.16006	0.505565
phase_162	162	0.711488	1.01942	0.228596	0.290414	0.404844	0.20839
phase_240	240	0.519536	0.62369	0.436112	0.344008	0.391284	0.311603

Panel 2 of 2: GBM, HGBM, XGBM, LGBM, ELM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM	ELM
phase_1	1	0.00193875	0.00217747	0.00234852	0.00261033	0.00269276
phase_3	3	0.0208532	0.0285098	0.0317376	0.045093	0.0662808
phase_39	39	0.0338847	0.033155	0.0466152	0.077429	0.0722836
phase_40	40	0.0347052	0.0470393	0.0602509	0.0640331	0.065352
phase_78	78	0.0649534	0.0900869	0.132517	0.144583	0.147244
phase_81	81	0.0690813	0.0779747	0.111915	0.149051	0.15427
phase_156	156	0.680302	0.884828	1.06937	1.22009	1.74182
phase_162	162	0.304907	0.441184	0.44421	0.632576	1.02676
phase_240	240	0.374422	0.41455	0.438863	0.476992	0.633718

Table 5 Equivalent - Phase RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00397461	0.0367435	0.00259315	0.00343561	0.00333782	0.00307895

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_3	3	0.0506014	0.0860856	0.0282609	0.0507192	0.0579582	0.0289877
phase_39	39	0.0614727	0.127013	0.0360729	0.0636581	0.0633639	0.0522371
phase_40	40	0.131569	0.127158	0.0529926	0.0735106	0.0982918	0.0507968
phase_78	78	0.332271	0.164189	0.211336	0.160754	0.168641	0.207215
phase_81	81	0.218491	0.225802	0.0825679	0.134366	0.13214	0.0622622
phase_156	156	2.41283	2.0493	1.06046	1.5605	2.09517	0.924908
phase_162	162	1.53987	1.46987	0.647753	0.769901	1.25598	0.708327
phase_240	240	1.00731	1.01822	0.940516	0.712705	0.90783	0.845185

Panel 2 of 2: GBM, HGBM, XGBM, LGBM, ELM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM	ELM
phase_1	1	0.00273946	0.00307036	0.00330225	0.00362244	0.00356586
phase_3	3	0.027172	0.0373394	0.0399144	0.0535386	0.0830624
phase_39	39	0.0482154	0.0569335	0.0736494	0.10155	0.0971095
phase_40	40	0.048643	0.0714217	0.0857178	0.0871438	0.0922018
phase_78	78	0.146262	0.185849	0.201331	0.259432	0.201695
phase_81	81	0.0933574	0.113812	0.156961	0.191343	0.199579
phase_156	156	1.0587	1.32219	1.45745	1.54621	2.08167
phase_162	162	0.757983	0.923536	0.833064	1.02708	1.53247
phase_240	240	0.93035	0.884716	0.819865	0.950281	0.976902

Best-Cell Selection Ledger

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
ampl_0	0	RF	0.0029211	SVR	0.00338731	RF (RMSE check: SVR)
ampl_1	1	ERT	2.99936e-05	HGBM	3.92211e-05	ERT (RMSE check: HGBM)
ampl_3	3	ERT	1.90726e-05	ERT	2.5487e-05	ERT
ampl_39	39	ERT	2.58271e-05	ERT	3.32522e-05	ERT
ampl_40	40	ERT	2.06984e-05	ERT	2.74003e-05	ERT
ampl_78	78	ERT	3.40623e-05	ERT	4.85815e-05	ERT
ampl_81	81	ERT	1.13607e-05	ERT	1.56353e-05	ERT
ampl_156	156	ERT	5.42847e-05	ERT	0.000226686	ERT
ampl_162	162	ERT	8.02514e-05	ET	0.000307436	ERT (RMSE check: ET)
ampl_240	240	ERT	3.96275e-05	ERT	7.97047e-05	ERT
phase_1	1	RF	0.00191692	RF	0.00259315	RF
phase_3	3	GBM	0.0208532	GBM	0.027172	GBM
phase_39	39	ERT	0.0271126	RF	0.0360729	ERT (RMSE check: RF)
phase_40	40	ERT	0.0327383	GBM	0.048643	ERT (RMSE check: GBM)
phase_78	78	GBM	0.0649534	GBM	0.146262	GBM
phase_81	81	ERT	0.0456165	ERT	0.0622622	ERT

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
phase_156	156	ERT	0.505565	ERT	0.924908	ERT
phase_162	162	ERT	0.20839	RF	0.647753	ERT (RMSE check: RF)
phase_240	240	ERT	0.311603	DT	0.712705	ERT (RMSE check: DT)

Backward Surface

Family columns: SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM, ELM .

Table 2 Equivalent - Amplitude MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.00429719	0.0187638	0.00326128	0.00362216	0.00323091	0.00346948
ampl_1	1	7.21116e-05	0.0180434	2.12857e-05	2.65706e-05	2.79374e-05	2.06273e-05
ampl_3	3	6.26989e-05	0.0142052	2.36905e-05	2.42053e-05	3.6441e-05	2.62655e-05
ampl_39	39	0.000101174	0.0222598	2.28753e-05	2.01634e-05	2.89436e-05	1.76074e-05
ampl_40	40	4.02579e-05	0.0259821	2.18196e-05	2.57958e-05	3.0195e-05	2.412e-05
ampl_78	78	0.00013487	0.0175306	5.50349e-05	5.25319e-05	0.000126116	3.64268e-05
ampl_81	81	2.13117e-05	0.0223986	1.07872e-05	1.14882e-05	1.9469e-05	8.76046e-06
ampl_156	156	0.000718831	0.0350925	0.000260409	0.000132202	0.000373914	0.000109416
ampl_162	162	0.000751263	0.0281716	0.00019246	0.000109396	0.000196806	8.52257e-05
ampl_240	240	0.0005152	0.024987	0.000178183	0.000120005	0.000183825	0.000135207

Panel 2 of 2: GBM, HGBM, XGBM, LGBM, ELM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM	ELM
ampl_0	0	0.00302314	0.00378368	0.004586	0.00245033	0.00739232
ampl_1	1	2.0832e-05	2.91305e-05	6.52717e-05	2.25836e-05	3.01773e-05
ampl_3	3	1.90749e-05	2.76483e-05	7.24577e-05	2.0776e-05	5.92126e-05
ampl_39	39	1.94171e-05	2.06079e-05	7.90442e-05	1.6854e-05	5.49178e-05
ampl_40	40	2.20437e-05	2.60511e-05	3.96378e-05	2.48788e-05	3.63881e-05
ampl_78	78	4.97133e-05	7.61737e-05	0.000145479	2.6598e-05	0.000280435
ampl_81	81	1.02443e-05	1.26117e-05	2.76087e-05	9.64368e-06	2.07512e-05
ampl_156	156	0.000274177	0.000284131	0.000408884	0.000290743	0.00104357
ampl_162	162	0.000153256	0.000338821	0.00028056	9.43304e-05	0.00104482
ampl_240	240	0.000174127	0.000269172	0.0003725	0.000116581	0.00044025

Table 3 Equivalent - Amplitude RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.00624162	0.0329994	0.00388728	0.00536033	0.00396137	0.00414577
ampl_1	1	0.000100359	0.0290015	3.17179e-05	4.02372e-05	3.80118e-05	2.92237e-05
ampl_3	3	8.0137e-05	0.0267382	3.21038e-05	3.67906e-05	4.36729e-05	3.50008e-05
ampl_39	39	0.000121816	0.0368861	2.90596e-05	2.77918e-05	3.85289e-05	2.37344e-05
ampl_40	40	5.14725e-05	0.0464764	2.95874e-05	3.65537e-05	4.1416e-05	3.24177e-05
ampl_78	78	0.000165646	0.0387349	7.49419e-05	7.35572e-05	0.000157222	5.1696e-05
ampl_81	81	2.79301e-05	0.0349958	1.53213e-05	1.64362e-05	2.88975e-05	1.26104e-05
ampl_156	156	0.00159816	0.0685187	0.000994547	0.000461825	0.0010175	0.000432476
ampl_162	162	0.00247527	0.0433351	0.00054455	0.000574387	0.000604509	0.000370452
ampl_240	240	0.00143899	0.0376344	0.000523307	0.000355973	0.000388673	0.000485608

Panel 2 of 2: GBM, HGBM, XGBM, LGBM, ELM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM	ELM
ampl_0	0	0.00368222	0.00429035	0.00542501	0.00322629	0.00974713
ampl_1	1	2.94098e-05	4.75846e-05	0.000108672	3.28906e-05	4.4137e-05
ampl_3	3	2.55762e-05	3.84266e-05	9.3682e-05	2.92663e-05	7.84404e-05
ampl_39	39	2.56806e-05	2.73323e-05	9.63518e-05	2.23516e-05	7.36454e-05
ampl_40	40	3.02315e-05	3.36259e-05	4.96259e-05	3.40799e-05	4.71594e-05
ampl_78	78	6.41438e-05	9.68442e-05	0.000194315	3.59238e-05	0.000372533
ampl_81	81	1.51924e-05	1.71525e-05	3.34652e-05	1.37455e-05	2.70445e-05
ampl_156	156	0.00120215	0.000819717	0.00104628	0.00099465	0.00173176
ampl_162	162	0.000406253	0.000840448	0.000698929	0.000232247	0.00239836
ampl_240	240	0.000532606	0.000875059	0.00103448	0.000225442	0.00128115

Table 4 Equivalent - Phase MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00267961	0.0250021	0.00151443	0.00184067	0.00224629	0.00158011
phase_3	3	0.0289354	0.085401	0.0227093	0.024766	0.0307531	0.0236435
phase_39	39	1.73853	1.46551	0.36563	0.203102	0.293192	0.243882
phase_40	40	0.226484	0.19159	0.093876	0.101007	0.131714	0.0790123
phase_78	78	0.153255	0.158349	0.0500226	0.0523841	0.0801005	0.0553777
phase_81	81	0.171261	0.221734	0.0804635	0.0912207	0.12357	0.0804235
phase_156	156	0.572634	0.636549	0.180663	0.120792	0.278798	0.106373
phase_162	162	0.746112	0.94923	0.100225	0.0532046	0.166394	0.0626116
phase_240	240	0.715218	0.708315	0.322211	0.168514	0.520119	0.286918

Panel 2 of 2: GBM, HGBM, XGBM, LGBM, ELM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM	ELM
phase_1	1	0.0015294	0.00187119	0.00210362	0.00174038	0.0029446
phase_3	3	0.0226183	0.0301106	0.0463593	0.0180247	0.0801661
phase_39	39	0.336998	0.500873	0.822733	0.368428	1.68169
phase_40	40	0.0978393	0.112472	0.158859	0.0975133	0.176486
phase_78	78	0.0455842	0.0858941	0.104748	0.0469313	0.130431
phase_81	81	0.0870526	0.110244	0.113511	0.0814266	0.179679
phase_156	156	0.163144	0.240075	0.26358	0.179082	0.623933
phase_162	162	0.0940569	0.173083	0.162021	0.0927631	0.874472
phase_240	240	0.29418	0.405538	0.482149	0.384359	0.750818

Table 5 Equivalent - Phase RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00384401	0.0377578	0.00206574	0.00254115	0.00348736	0.00213017

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_3	3	0.0411088	0.109135	0.0314359	0.0341871	0.0414338	0.0356001
phase_39	39	2.12783	1.8702	0.958068	0.82471	1.21723	0.849285
phase_40	40	0.3468	0.296805	0.128839	0.149163	0.180527	0.107083
phase_78	78	0.350233	0.213038	0.0953936	0.0880881	0.138744	0.124823
phase_81	81	0.2245	0.290191	0.118077	0.135455	0.16677	0.125367
phase_156	156	1.08322	0.96245	0.441122	0.641618	0.651211	0.334748
phase_162	162	1.44516	1.29006	0.228194	0.0809374	0.58247	0.133295
phase_240	240	1.3595	1.17318	0.740668	0.590811	1.12751	0.743524

Panel 2 of 2: GBM, HGBM, XGBM, LGBM, ELM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM	ELM
phase_1	1	0.00209734	0.00299787	0.00314551	0.00233636	0.00393481
phase_3	3	0.0294579	0.0375849	0.05994	0.0272988	0.104573
phase_39	39	0.94203	0.954013	1.18958	0.938005	2.19552
phase_40	40	0.132488	0.160093	0.259128	0.12908	0.276721
phase_78	78	0.0837849	0.182508	0.224614	0.0716468	0.169502
phase_81	81	0.126905	0.159944	0.15135	0.105663	0.236343
phase_156	156	0.424279	0.538131	0.559001	0.458084	0.976768
phase_162	162	0.213448	0.292846	0.311383	0.145162	1.24462
phase_240	240	0.647993	0.901278	0.863736	0.772565	1.1873

Best-Cell Selection Ledger

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
ampl_0	0	LGBM	0.00245033	LGBM	0.00322629	LGBM
ampl_1	1	ERT	2.06273e-05	ERT	2.92237e-05	ERT
ampl_3	3	GBM	1.90749e-05	GBM	2.55762e-05	GBM
ampl_39	39	LGBM	1.6854e-05	LGBM	2.23516e-05	LGBM
ampl_40	40	RF	2.18196e-05	RF	2.95874e-05	RF
ampl_78	78	LGBM	2.6598e-05	LGBM	3.59238e-05	LGBM
ampl_81	81	ERT	8.76046e-06	ERT	1.26104e-05	ERT
ampl_156	156	ERT	0.000109416	ERT	0.000432476	ERT
ampl_162	162	ERT	8.52257e-05	LGBM	0.000232247	ERT (RMSE check: LGBM)
ampl_240	240	LGBM	0.000116581	LGBM	0.000225442	LGBM
phase_1	1	RF	0.00151443	RF	0.00206574	RF
phase_3	3	LGBM	0.0180247	LGBM	0.0272988	LGBM
phase_39	39	DT	0.203102	DT	0.82471	DT
phase_40	40	ERT	0.0790123	ERT	0.107083	ERT
phase_78	78	GBM	0.0455842	LGBM	0.0716468	GBM (RMSE check: LGBM)
phase_81	81	ERT	0.0804235	LGBM	0.105663	ERT (RMSE check: LGBM)
phase_156	156	ERT	0.106373	ERT	0.334748	ERT

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
phase_162	162	DT	0.0532046	DT	0.0809374	DT
phase_240	240	DT	0.168514	DT	0.590811	DT

Polished Dataset + Setpoints

- Dataset: polished_dataset
- Input mode: setpoints
- Metric rows loaded: 570
- Source mode: validation_summary

Forward Surface

Family columns: SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM .

Table 2 Equivalent - Amplitude MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.000748564	0.0294033	0.00132913	0.000920489	0.000991	0.000769024
ampl_1	1	5.4081e-05	0.0232739	2.57464e-05	2.92126e-05	3.03177e-05	2.7015e-05
ampl_3	3	0.000145381	0.0252054	2.05147e-05	2.13643e-05	2.19543e-05	2.14493e-05
ampl_39	39	0.000172652	0.023849	3.08568e-05	4.11719e-05	4.01066e-05	3.19989e-05
ampl_40	40	8.44586e-05	0.0339631	2.6672e-05	3.10544e-05	4.14545e-05	2.77176e-05
ampl_78	78	0.00015222	0.0321545	3.92943e-05	6.59289e-05	6.09289e-05	3.9163e-05
ampl_81	81	9.00347e-05	0.0268317	1.14703e-05	1.38479e-05	1.24472e-05	1.05736e-05
ampl_156	156	0.000356183	0.0338399	8.37181e-05	5.7261e-05	7.98961e-05	4.10022e-05
ampl_162	162	0.000469851	0.0323956	8.52018e-05	6.41517e-05	5.73224e-05	2.7079e-05
ampl_240	240	0.000305713	0.0204332	5.27454e-05	3.95193e-05	4.7756e-05	3.25162e-05

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.000992455	0.00191049	0.00162904	0.00648623
ampl_1	1	2.28739e-05	2.57732e-05	5.29963e-05	3.25939e-05
ampl_3	3	1.92887e-05	2.54849e-05	7.87067e-05	5.40891e-05
ampl_39	39	3.12746e-05	3.833e-05	0.000104187	0.000100191
ampl_40	40	2.65085e-05	3.27083e-05	7.30745e-05	4.41878e-05
ampl_78	78	3.94912e-05	7.74787e-05	0.000127797	0.000213713
ampl_81	81	1.12049e-05	1.41602e-05	5.09462e-05	2.55819e-05
ampl_156	156	7.79321e-05	0.000176057	0.000201092	0.000326268
ampl_162	162	7.90026e-05	0.000196612	0.000208811	0.000366331
ampl_240	240	5.49509e-05	5.54731e-05	0.000141483	0.000113788

Table 3 Equivalent - Amplitude RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.00107841	0.0557488	0.00174811	0.00150106	0.00136872	0.00105512
ampl_1	1	7.03725e-05	0.0505225	3.22457e-05	3.75857e-05	3.96282e-05	3.46878e-05
ampl_3	3	0.000167017	0.0540432	2.65922e-05	3.0559e-05	2.96112e-05	2.89688e-05
ampl_39	39	0.000202311	0.0565217	4.29117e-05	6.4853e-05	5.59825e-05	4.64667e-05
ampl_40	40	9.92476e-05	0.066584	4.03538e-05	4.8403e-05	7.13741e-05	4.20016e-05
ampl_78	78	0.00018438	0.0668506	5.64342e-05	9.44923e-05	8.59642e-05	5.79877e-05
ampl_81	81	0.000101748	0.0580985	1.68141e-05	1.98414e-05	1.84426e-05	1.57922e-05
ampl_156	156	0.000763936	0.0664826	0.000289051	0.000187919	0.000262648	0.000133233
ampl_162	162	0.00150698	0.0660267	0.000262637	0.000204199	0.000165767	8.00065e-05
ampl_240	240	0.000465864	0.0319315	0.000118049	5.82884e-05	6.87839e-05	6.00519e-05

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00126774	0.00236884	0.00246666	0.00769967
ampl_1	1	2.91489e-05	3.30839e-05	6.79555e-05	4.10576e-05
ampl_3	3	2.55216e-05	3.28466e-05	0.00010161	6.68492e-05
ampl_39	39	4.42077e-05	5.0013e-05	0.000133321	0.000118642
ampl_40	40	4.13792e-05	4.73766e-05	9.47978e-05	6.13187e-05
ampl_78	78	5.31953e-05	0.000103314	0.000168216	0.000251128
ampl_81	81	1.68186e-05	1.94124e-05	6.43327e-05	3.3026e-05
ampl_156	156	0.000228112	0.000469723	0.000544465	0.0006729
ampl_162	162	0.000268125	0.000612269	0.000614859	0.000957035
ampl_240	240	0.00012478	0.000113387	0.000231143	0.000203018

Table 4 Equivalent - Phase MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.0025097	0.013746	0.00165038	0.00194854	0.0021261	0.00195443
phase_3	3	0.0336917	0.0727522	0.0204178	0.0263402	0.0301677	0.0244851
phase_39	39	0.0253174	0.0998355	0.0211707	0.0229524	0.0233407	0.0189912
phase_40	40	0.0503443	0.0822119	0.0417468	0.0508716	0.0515189	0.0429527
phase_78	78	0.15959	0.181007	0.0831793	0.105963	0.0761081	0.0898032
phase_81	81	0.0834078	0.156329	0.0481367	0.0627761	0.0563076	0.0453614
phase_156	156	1.07536	1.29151	0.582387	0.497722	0.481487	0.359726
phase_162	162	0.49565	0.75881	0.281999	0.254107	0.2004	0.207715
phase_240	240	0.766827	0.756654	0.325664	0.145582	0.300122	0.242563

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00181678	0.00187719	0.00181933	0.00244227
phase_3	3	0.0210674	0.030984	0.0257969	0.0530085
phase_39	39	0.0182133	0.0319866	0.0292705	0.0789214
phase_40	40	0.0400469	0.0469295	0.0506291	0.0625339
phase_78	78	0.0931014	0.11669	0.127938	0.168479
phase_81	81	0.0473044	0.0632147	0.0670389	0.142364
phase_156	156	0.562997	0.721864	0.687853	0.943033
phase_162	162	0.316012	0.355762	0.343733	0.582623
phase_240	240	0.247804	0.506181	0.440873	0.64463

Table 5 Equivalent - Phase RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00369658	0.031276	0.00230991	0.0024745	0.00280103	0.00280993

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_3	3	0.0425176	0.0918986	0.0283452	0.0345489	0.0376841	0.0322408
phase_39	39	0.0362159	0.133566	0.0327849	0.0308022	0.0321086	0.0265687
phase_40	40	0.0801249	0.108856	0.0670098	0.0779802	0.0707268	0.0668143
phase_78	78	0.27668	0.298146	0.237482	0.39107	0.175385	0.288322
phase_81	81	0.109969	0.204298	0.0670047	0.082113	0.0732442	0.0649923
phase_156	156	1.72625	1.84219	1.19293	1.48855	1.42803	0.988177
phase_162	162	1.15157	1.22488	0.833156	1.04966	0.788483	0.694904
phase_240	240	1.60209	1.26225	0.727011	0.635436	0.954825	0.637673

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00261915	0.0027905	0.00270348	0.00339916
phase_3	3	0.0290049	0.0401291	0.0365085	0.0657637
phase_39	39	0.0266143	0.044045	0.0473758	0.101829
phase_40	40	0.062705	0.0715206	0.0775755	0.0898916
phase_78	78	0.305955	0.260769	0.293856	0.340729
phase_81	81	0.0660989	0.086626	0.0906761	0.173738
phase_156	156	1.24559	1.23859	1.17765	1.39191
phase_162	162	0.876285	0.832763	0.822323	0.975679
phase_240	240	0.609337	1.04985	0.900178	1.25853

Best-Cell Selection Ledger

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
ampl_0	0	SVR	0.000748564	ERT	0.00105512	SVR (RMSE check: ERT)
ampl_1	1	GBM	2.28739e-05	GBM	2.91489e-05	GBM
ampl_3	3	GBM	1.92887e-05	GBM	2.55216e-05	GBM
ampl_39	39	RF	3.08568e-05	RF	4.29117e-05	RF
ampl_40	40	GBM	2.65085e-05	RF	4.03538e-05	GBM (RMSE check: RF)
ampl_78	78	ERT	3.9163e-05	GBM	5.31953e-05	ERT (RMSE check: GBM)
ampl_81	81	ERT	1.05736e-05	ERT	1.57922e-05	ERT
ampl_156	156	ERT	4.10022e-05	ERT	0.000133233	ERT
ampl_162	162	ERT	2.7079e-05	ERT	8.00065e-05	ERT
ampl_240	240	ERT	3.25162e-05	DT	5.82884e-05	ERT (RMSE check: DT)
phase_1	1	RF	0.00165038	RF	0.00230991	RF
phase_3	3	RF	0.0204178	RF	0.0283452	RF
phase_39	39	GBM	0.0182133	ERT	0.0265687	GBM (RMSE check: ERT)
phase_40	40	GBM	0.0400469	GBM	0.062705	GBM
phase_78	78	ET	0.0761081	ET	0.175385	ET
phase_81	81	ERT	0.0453614	ERT	0.0649923	ERT

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
phase_156	156	ERT	0.359726	ERT	0.988177	ERT
phase_162	162	ET	0.2004	ERT	0.694904	ET (RMSE check: ERT)
phase_240	240	DT	0.145582	GBM	0.609337	DT (RMSE check: GBM)

Backward Surface

Family columns: SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM .

Table 2 Equivalent - Amplitude MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.000878143	0.0212593	0.0013777	0.00227103	0.000723022	0.000695714
ampl_1	1	0.000177126	0.0217882	2.70184e-05	2.75766e-05	2.87226e-05	2.25195e-05
ampl_3	3	0.000131764	0.0175809	2.41663e-05	3.59769e-05	2.72481e-05	2.25279e-05
ampl_39	39	9.29819e-05	0.0220875	2.02536e-05	2.94057e-05	2.59767e-05	2.15566e-05
ampl_40	40	4.71358e-05	0.0191971	2.88667e-05	3.31922e-05	4.00967e-05	2.99197e-05
ampl_78	78	0.000227138	0.0182973	5.97498e-05	9.89256e-05	6.95992e-05	3.96771e-05
ampl_81	81	4.24634e-05	0.0200997	1.08778e-05	1.48488e-05	1.24414e-05	9.30678e-06
ampl_156	156	0.000597941	0.0221492	0.000192254	0.000222675	9.67728e-05	9.19443e-05
ampl_162	162	0.000500101	0.0184939	0.000197114	0.000103617	5.6251e-05	3.05272e-05
ampl_240	240	0.000335452	0.0146316	0.000162647	0.000127615	8.51129e-05	8.54199e-05

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00237412	0.00187147	0.0018695	0.000623818
ampl_1	1	2.75321e-05	2.76878e-05	6.12132e-05	2.57323e-05
ampl_3	3	2.97891e-05	2.77415e-05	8.3327e-05	2.04977e-05
ampl_39	39	2.13804e-05	2.50807e-05	8.4512e-05	1.94426e-05
ampl_40	40	3.12774e-05	3.07425e-05	4.83798e-05	2.99887e-05
ampl_78	78	9.0879e-05	7.44539e-05	0.00013382	2.77967e-05
ampl_81	81	9.99874e-06	1.32954e-05	2.72327e-05	9.48381e-06
ampl_156	156	0.000243619	0.000302913	0.000332984	0.000231125
ampl_162	162	0.000184607	0.00026871	0.000213247	0.000118476
ampl_240	240	0.000168594	0.000222699	0.000247886	0.000105154

Table 3 Equivalent - Amplitude RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.00124141	0.0432267	0.00192752	0.00275406	0.000984209	0.00100565
ampl_1	1	0.000188894	0.0572224	4.06142e-05	3.64e-05	3.91862e-05	3.0293e-05
ampl_3	3	0.00015476	0.0475276	3.82057e-05	5.04829e-05	4.4711e-05	3.60138e-05
ampl_39	39	0.000108285	0.0493071	2.98467e-05	4.08865e-05	4.17453e-05	3.34854e-05
ampl_40	40	5.84945e-05	0.0462838	4.21788e-05	4.72622e-05	5.79503e-05	4.30555e-05
ampl_78	78	0.000262946	0.0493442	8.39832e-05	0.000127544	9.57646e-05	5.11281e-05
ampl_81	81	4.69126e-05	0.0438215	1.83586e-05	2.55154e-05	2.12023e-05	1.66222e-05
ampl_156	156	0.0021765	0.0520806	0.000725249	0.000667422	0.000495234	0.000342015
ampl_162	162	0.00148596	0.0448979	0.000589234	0.000227782	0.000204337	8.89209e-05
ampl_240	240	0.000996253	0.0289771	0.00052279	0.000215076	0.00020448	0.000215657

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00286507	0.00236574	0.00257443	0.000812979
ampl_1	1	3.8401e-05	3.85516e-05	9.82192e-05	3.52964e-05
ampl_3	3	4.90406e-05	4.41965e-05	0.000120517	3.12569e-05
ampl_39	39	3.05721e-05	3.71332e-05	0.00010196	2.89869e-05
ampl_40	40	4.52546e-05	4.36316e-05	5.86742e-05	4.29992e-05
ampl_78	78	0.000109218	9.31098e-05	0.000171255	3.81085e-05
ampl_81	81	1.66692e-05	2.10659e-05	3.73935e-05	1.49736e-05
ampl_156	156	0.000630191	0.00100674	0.00111236	0.000580753
ampl_162	162	0.000477327	0.000721602	0.000598474	0.00029402
ampl_240	240	0.00054354	0.00067227	0.000674817	0.000267024

Table 4 Equivalent - Phase MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00268537	0.0215304	0.00164448	0.00224398	0.00194629	0.00147046
phase_3	3	0.0255561	0.0979003	0.0263511	0.0322145	0.0275972	0.0221575
phase_39	39	0.850215	1.23679	0.326963	0.231609	0.237228	0.250191
phase_40	40	0.181409	0.256653	0.107192	0.125405	0.132724	0.0870801
phase_78	78	0.124086	0.127039	0.0783109	0.0790418	0.0595767	0.0701809
phase_81	81	0.139894	0.177046	0.0829331	0.119519	0.111857	0.0707507
phase_156	156	0.392967	0.545313	0.153452	0.152396	0.191639	0.120446
phase_162	162	0.311631	0.576634	0.0671225	0.0943387	0.0511645	0.0453581
phase_240	240	0.373277	0.635258	0.320211	0.225827	0.138411	0.199767

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00157792	0.00194924	0.00212909	0.00170311
phase_3	3	0.0363193	0.0316653	0.0268202	0.0181157
phase_39	39	0.408832	0.492335	0.516416	0.3689
phase_40	40	0.100007	0.143129	0.155347	0.101143
phase_78	78	0.0675646	0.0761916	0.0926336	0.0579179
phase_81	81	0.0822115	0.100723	0.0958313	0.0800544
phase_156	156	0.175931	0.206933	0.191042	0.138619
phase_162	162	0.135332	0.130277	0.12271	0.0431247
phase_240	240	0.238079	0.465994	0.36514	0.364304

Table 5 Equivalent - Phase RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00440989	0.0439222	0.00222792	0.00412737	0.00258861	0.0019963

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_3	3	0.0369234	0.122374	0.0357796	0.0415183	0.037187	0.0292334
phase_39	39	1.36871	1.7484	0.844381	1.07395	1.08098	0.847479
phase_40	40	0.331575	0.370072	0.171222	0.20465	0.268756	0.142841
phase_78	78	0.25784	0.180765	0.198648	0.137739	0.11379	0.184209
phase_81	81	0.177273	0.215674	0.106024	0.162995	0.154996	0.0923238
phase_156	156	0.832814	0.903346	0.369552	0.457737	0.659514	0.455612
phase_162	162	0.591693	0.889113	0.108931	0.156047	0.0775683	0.0846649
phase_240	240	0.971372	1.04653	0.624273	0.570448	0.450707	0.477846

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00205313	0.00293993	0.00298142	0.00229031
phase_3	3	0.0450889	0.0403717	0.0372678	0.024326
phase_39	39	0.697028	0.972275	1.04969	0.828096
phase_40	40	0.175631	0.221033	0.245185	0.192371
phase_78	78	0.156399	0.161476	0.192463	0.126814
phase_81	81	0.103397	0.130001	0.121514	0.103419
phase_156	156	0.347004	0.432992	0.380866	0.2781
phase_162	162	0.192957	0.193097	0.197834	0.0739893
phase_240	240	0.420362	0.892726	0.715874	0.61474

Best-Cell Selection Ledger

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
ampl_0	0	LGBM	0.000623818	LGBM	0.000812979	LGBM
ampl_1	1	ERT	2.25195e-05	ERT	3.0293e-05	ERT
ampl_3	3	LGBM	2.04977e-05	LGBM	3.12569e-05	LGBM
ampl_39	39	LGBM	1.94426e-05	LGBM	2.89869e-05	LGBM
ampl_40	40	RF	2.88667e-05	RF	4.21788e-05	RF
ampl_78	78	LGBM	2.77967e-05	LGBM	3.81085e-05	LGBM
ampl_81	81	ERT	9.30678e-06	LGBM	1.49736e-05	ERT (RMSE check: LGBM)
ampl_156	156	ERT	9.19443e-05	ERT	0.000342015	ERT
ampl_162	162	ERT	3.05272e-05	ERT	8.89209e-05	ERT
ampl_240	240	ET	8.51129e-05	ET	0.00020448	ET
phase_1	1	ERT	0.00147046	ERT	0.0019963	ERT
phase_3	3	LGBM	0.0181157	LGBM	0.024326	LGBM
phase_39	39	DT	0.231609	GBM	0.697028	DT (RMSE check: GBM)
phase_40	40	ERT	0.0870801	ERT	0.142841	ERT
phase_78	78	LGBM	0.0579179	ET	0.11379	LGBM (RMSE check: ET)
phase_81	81	ERT	0.0707507	ERT	0.0923238	ERT
phase_156	156	ERT	0.120446	LGBM	0.2781	ERT (RMSE check: LGBM)

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
phase_162	162	LGBM	0.0431247	LGBM	0.0739893	LGBM
phase_240	240	ET	0.138411	GBM	0.420362	ET (RMSE check: GBM)

Global Surface

Family columns: SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM .

Table 2 Equivalent - Amplitude MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.0156559	0.0203548	0.000917173	0.00108796	0.00115956	0.000784491
ampl_1	1	0.000172263	0.00802753	2.66196e-05	3.25522e-05	3.1181e-05	2.56115e-05
ampl_3	3	0.000169671	0.00653182	2.04705e-05	2.27158e-05	2.31351e-05	2.07127e-05
ampl_39	39	0.000210865	0.00817687	2.17073e-05	3.15443e-05	2.73856e-05	2.26208e-05
ampl_40	40	0.000251705	0.00553398	2.66152e-05	2.86737e-05	3.06929e-05	2.50924e-05
ampl_78	78	0.000222833	0.0107326	3.76979e-05	5.28057e-05	5.55613e-05	3.7709e-05
ampl_81	81	0.000120663	0.0071898	1.07739e-05	1.28853e-05	1.38757e-05	1.11178e-05
ampl_156	156	0.000688856	0.0111127	0.000139415	0.000138055	0.00013822	9.65902e-05
ampl_162	162	0.000526237	0.00917815	6.20027e-05	6.25726e-05	7.02969e-05	4.53373e-05
ampl_240	240	0.000357056	0.00744795	9.85699e-05	0.000108632	0.000148384	0.000136539

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00267243	0.00110819	0.00112555	0.000992191
ampl_1	1	2.98505e-05	2.72457e-05	5.44221e-05	2.60118e-05
ampl_3	3	2.47808e-05	2.02009e-05	6.36521e-05	1.93222e-05
ampl_39	39	4.41559e-05	2.36618e-05	8.85125e-05	2.12985e-05
ampl_40	40	3.99205e-05	2.76882e-05	6.43255e-05	2.51697e-05
ampl_78	78	8.04946e-05	4.56745e-05	0.000106457	3.01252e-05
ampl_81	81	1.75155e-05	1.3117e-05	4.83386e-05	1.10394e-05
ampl_156	156	0.000213871	0.000223984	0.000290866	0.000190973
ampl_162	162	0.000139214	7.30669e-05	0.000144719	8.79801e-05
ampl_240	240	8.80481e-05	0.000157348	0.00021781	0.000133826

Table 3 Equivalent - Amplitude RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.0191204	0.0267557	0.00215482	0.00251192	0.00221109	0.00191334
ampl_1	1	0.000187927	0.0166662	3.82955e-05	4.41147e-05	4.49504e-05	3.73956e-05
ampl_3	3	0.000205732	0.0128392	3.00682e-05	3.38024e-05	3.70774e-05	2.9912e-05
ampl_39	39	0.000255756	0.0156072	2.99623e-05	4.39118e-05	3.91648e-05	3.24746e-05
ampl_40	40	0.000264557	0.0122896	3.90823e-05	3.97988e-05	4.27685e-05	3.73538e-05
ampl_78	78	0.000278071	0.0190911	5.44844e-05	7.21474e-05	7.55617e-05	5.24132e-05
ampl_81	81	0.000145078	0.0162407	1.83861e-05	2.05647e-05	2.12414e-05	1.79783e-05
ampl_156	156	0.0022958	0.0226072	0.000559498	0.000473122	0.000404718	0.00031975
ampl_162	162	0.00175462	0.0175956	0.000202081	0.000170358	0.00019439	0.00015848
ampl_240	240	0.000864756	0.015796	0.00025344	0.000324476	0.000499917	0.000338261

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00359339	0.00235823	0.00240993	0.0024809
ampl_1	1	4.11085e-05	4.03786e-05	7.24916e-05	3.81387e-05
ampl_3	3	3.48772e-05	2.8546e-05	8.1721e-05	2.85489e-05
ampl_39	39	5.59724e-05	3.30474e-05	0.000116431	3.02244e-05
ampl_40	40	4.98711e-05	3.9748e-05	8.44289e-05	3.66856e-05
ampl_78	78	9.92558e-05	6.31463e-05	0.000142352	4.25608e-05
ampl_81	81	2.41387e-05	2.1026e-05	6.4229e-05	1.74133e-05
ampl_156	156	0.00061781	0.000913716	0.000909519	0.000506144
ampl_162	162	0.000338935	0.00016564	0.00033997	0.000209959
ampl_240	240	0.000210591	0.000393091	0.000551578	0.000315799

Table 4 Equivalent - Phase MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.0101759	0.0155415	0.0020719	0.00268127	0.00231501	0.00227012
phase_3	3	0.226685	0.229007	0.0201524	0.0250584	0.0264927	0.0218246
phase_39	39	0.845315	1.09345	0.114402	0.133051	0.097663	0.103757
phase_40	40	0.192786	0.22048	0.0661915	0.0760454	0.0714671	0.065664
phase_78	78	0.396387	0.408786	0.0540032	0.0678423	0.0755244	0.051397
phase_81	81	0.997655	0.974842	0.0592676	0.0777414	0.0769398	0.0596714
phase_156	156	1.02004	1.17939	0.34019	0.37524	0.385694	0.307435
phase_162	162	0.425469	0.800536	0.0849107	0.0949722	0.145188	0.0819587
phase_240	240	2.02113	2.09429	0.311247	0.328083	0.511719	0.308097

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.0025628	0.00227524	0.00203238	0.00186893
phase_3	3	0.0333731	0.024645	0.0216629	0.0198815
phase_39	39	0.203418	0.154774	0.162164	0.184635
phase_40	40	0.0717133	0.0702019	0.074052	0.0669926
phase_78	78	0.0885852	0.0768515	0.072187	0.0678489
phase_81	81	0.13614	0.0761886	0.0676996	0.0638208
phase_156	156	0.394425	0.450806	0.441404	0.495762
phase_162	162	0.188317	0.139504	0.112098	0.144505
phase_240	240	0.393299	0.487416	0.403316	0.464343

Table 5 Equivalent - Phase RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.0123098	0.020824	0.00330762	0.00468138	0.00367207	0.0034834

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_3	3	0.325041	0.284605	0.0276204	0.0379931	0.0385762	0.0317374
phase_39	39	1.32781	1.53107	0.505745	0.764419	0.477528	0.458963
phase_40	40	0.301887	0.309243	0.117868	0.130749	0.121862	0.118611
phase_78	78	0.50126	0.524407	0.121747	0.137497	0.168653	0.124472
phase_81	81	1.20977	1.05994	0.0839313	0.106699	0.113186	0.0902668
phase_156	156	1.48875	1.53569	0.877726	1.19647	0.861363	0.770493
phase_162	162	0.610809	1.12611	0.262415	0.35958	0.500324	0.232623
phase_240	240	2.46384	2.1841	0.997373	1.14765	1.31112	0.904627

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00404032	0.00329108	0.00278526	0.00252914
phase_3	3	0.0423266	0.0391159	0.0348155	0.0328484
phase_39	39	0.583647	0.475857	0.529911	0.543805
phase_40	40	0.125696	0.119567	0.125355	0.115733
phase_78	78	0.156195	0.165735	0.161552	0.143122
phase_81	81	0.162954	0.105482	0.09916	0.0931345
phase_156	156	1.00741	0.914904	0.881375	0.932239
phase_162	162	0.359983	0.309376	0.214857	0.272781
phase_240	240	0.91467	1.0633	0.966874	0.994939

Best-Cell Selection Ledger

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
ampl_0	0	ERT	0.000784491	ERT	0.00191334	ERT
ampl_1	1	ERT	2.56115e-05	ERT	3.73956e-05	ERT
ampl_3	3	LGBM	1.93222e-05	HGBM	2.8546e-05	LGBM (RMSE check: HGBM)
ampl_39	39	LGBM	2.12985e-05	RF	2.99623e-05	LGBM (RMSE check: RF)
ampl_40	40	ERT	2.50924e-05	LGBM	3.66856e-05	ERT (RMSE check: LGBM)
ampl_78	78	LGBM	3.01252e-05	LGBM	4.25608e-05	LGBM
ampl_81	81	RF	1.07739e-05	LGBM	1.74133e-05	RF (RMSE check: LGBM)
ampl_156	156	ERT	9.65902e-05	ERT	0.00031975	ERT
ampl_162	162	ERT	4.53373e-05	ERT	0.00015848	ERT
ampl_240	240	GBM	8.80481e-05	GBM	0.000210591	GBM
phase_1	1	LGBM	0.00186893	LGBM	0.00252914	LGBM
phase_3	3	LGBM	0.0198815	RF	0.0276204	LGBM (RMSE check: RF)
phase_39	39	ET	0.097663	ERT	0.458963	ET (RMSE check: ERT)
phase_40	40	ERT	0.065664	LGBM	0.115733	ERT (RMSE check: LGBM)
phase_78	78	ERT	0.051397	RF	0.121747	ERT (RMSE check: RF)

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
phase_81	81	RF	0.0592676	RF	0.0839313	RF
phase_156	156	ERT	0.307435	ERT	0.770493	ERT
phase_162	162	ERT	0.0819587	XGBM	0.214857	ERT (RMSE check: XGBM)
phase_240	240	ERT	0.308097	ERT	0.904627	ERT

Polished Dataset + Actual Values

- Dataset: polished_dataset
- Input mode: actual_values
- Metric rows loaded: 570
- Source mode: validation_summary

Forward Surface

Family columns: SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM .

Table 2 Equivalent - Amplitude MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.000718839	0.0202281	0.00130693	0.00139082	0.00108981	0.000808464
ampl_1	1	5.4081e-05	0.0290525	2.47708e-05	3.1738e-05	3.12766e-05	2.31427e-05
ampl_3	3	0.000145381	0.022325	1.99498e-05	2.52596e-05	2.49015e-05	1.98433e-05
ampl_39	39	0.000197129	0.0222592	2.96629e-05	5.38906e-05	4.99037e-05	3.06737e-05
ampl_40	40	8.44586e-05	0.0227821	2.75789e-05	3.72429e-05	4.00456e-05	2.54591e-05
ampl_78	78	0.000153046	0.0321306	5.32729e-05	6.28133e-05	7.05038e-05	4.70257e-05
ampl_81	81	9.00347e-05	0.0261786	1.17108e-05	1.40493e-05	1.61188e-05	1.12285e-05
ampl_156	156	0.000343646	0.0227924	0.000130857	8.76672e-05	0.000161826	6.15641e-05
ampl_162	162	0.000455859	0.0262349	0.000182552	6.08304e-05	7.04288e-05	4.15775e-05
ampl_240	240	0.000288224	0.0246914	6.47582e-05	4.90539e-05	5.91392e-05	3.40349e-05

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00126613	0.00205371	0.00204164	0.000751858
ampl_1	1	2.43518e-05	2.58716e-05	5.29963e-05	2.53772e-05
ampl_3	3	2.18818e-05	2.65947e-05	8.03141e-05	2.04023e-05
ampl_39	39	3.11135e-05	3.998e-05	0.000105918	2.67456e-05
ampl_40	40	2.94292e-05	3.24804e-05	7.30745e-05	3.02107e-05
ampl_78	78	5.18818e-05	8.57062e-05	0.000130228	3.68155e-05
ampl_81	81	1.14572e-05	1.43715e-05	5.09462e-05	1.04092e-05
ampl_156	156	0.000114335	0.000206002	0.00020442	0.000125314
ampl_162	162	0.000172532	0.000203292	0.000236852	0.000156842
ampl_240	240	7.94708e-05	7.46447e-05	0.00016647	9.51925e-05

Table 3 Equivalent - Amplitude RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.00104123	0.0352655	0.00175396	0.00311184	0.00154469	0.00115498
ampl_1	1	7.03725e-05	0.0368434	3.17193e-05	4.21933e-05	4.08831e-05	2.98514e-05
ampl_3	3	0.000167017	0.0334958	2.73983e-05	3.63794e-05	3.43315e-05	2.73845e-05
ampl_39	39	0.000225558	0.0336646	4.33871e-05	7.4857e-05	7.57432e-05	4.77076e-05
ampl_40	40	9.92476e-05	0.0343111	3.78382e-05	6.00553e-05	6.50539e-05	3.61431e-05
ampl_78	78	0.000188016	0.0433831	7.15845e-05	8.80811e-05	9.78815e-05	6.74232e-05
ampl_81	81	0.000101748	0.036541	1.69644e-05	1.98602e-05	2.33258e-05	1.54825e-05
ampl_156	156	0.000767559	0.0338169	0.00044745	0.000305714	0.0007616	0.000234491
ampl_162	162	0.00154793	0.0371746	0.000661113	0.00015795	0.000246151	0.00011706
ampl_240	240	0.000482149	0.0358373	0.000132739	7.07926e-05	0.000119373	5.3833e-05

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00168922	0.00256428	0.00293613	0.000988936
ampl_1	1	3.09964e-05	3.37361e-05	6.79555e-05	3.32426e-05
ampl_3	3	2.94392e-05	3.49586e-05	0.0001021	2.68918e-05
ampl_39	39	4.27296e-05	5.31952e-05	0.000134926	4.1021e-05
ampl_40	40	4.14926e-05	4.49906e-05	9.47978e-05	4.48026e-05
ampl_78	78	7.09721e-05	0.000110508	0.000172883	5.13891e-05
ampl_81	81	1.58057e-05	2.11856e-05	6.43327e-05	1.53275e-05
ampl_156	156	0.000330732	0.000573222	0.000525614	0.000360075
ampl_162	162	0.000762852	0.000647665	0.000680446	0.000539521
ampl_240	240	0.000226516	0.000225534	0.000330237	0.000297743

Table 4 Equivalent - Phase MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00247434	0.026175	0.00162887	0.00235042	0.00278845	0.00179615
phase_3	3	0.0340603	0.0913634	0.0261114	0.030708	0.0409561	0.0249129
phase_39	39	0.025169	0.0660206	0.0223827	0.0232558	0.0284039	0.0201729
phase_40	40	0.0504003	0.0693434	0.0447625	0.0510739	0.0576589	0.0411526
phase_78	78	0.159626	0.139742	0.0932301	0.107206	0.108892	0.09352
phase_81	81	0.0832659	0.138154	0.0540683	0.0746729	0.0741155	0.04529
phase_156	156	1.07502	1.36918	0.67695	0.63532	0.653339	0.471632
phase_162	162	0.495984	0.969732	0.257478	0.212884	0.272101	0.243755
phase_240	240	0.766222	0.855356	0.410153	0.164686	0.393957	0.351379

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00180807	0.00188766	0.00191935	0.00158896
phase_3	3	0.0245744	0.0327911	0.030218	0.0227335
phase_39	39	0.0213026	0.0326097	0.0317731	0.0175592
phase_40	40	0.0452036	0.0504163	0.0522935	0.0412382
phase_78	78	0.0899856	0.107512	0.119391	0.0758149
phase_81	81	0.0517666	0.0687608	0.0714134	0.0492309
phase_156	156	0.628382	0.739474	0.78646	0.656478
phase_162	162	0.260216	0.334499	0.35739	0.218499
phase_240	240	0.355237	0.549711	0.498655	0.415218

Table 5 Equivalent - Phase RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00358489	0.0358289	0.00224899	0.0030367	0.00413373	0.00255174

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_3	3	0.0426571	0.12082	0.0346515	0.0373525	0.0537245	0.0335425
phase_39	39	0.0360695	0.082434	0.0332546	0.0321677	0.0381631	0.0265764
phase_40	40	0.0799549	0.0920131	0.0661182	0.0793592	0.0903941	0.0608495
phase_78	78	0.276848	0.189362	0.229668	0.341413	0.233049	0.259441
phase_81	81	0.109887	0.175225	0.0758439	0.108574	0.0977402	0.0629514
phase_156	156	1.72697	1.99237	1.26035	1.66846	1.40627	0.96815
phase_162	162	1.1517	1.52093	0.744869	0.874995	0.988992	0.779322
phase_240	240	1.59473	1.3284	0.904678	0.634265	1.09484	0.750481

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00240062	0.00271213	0.0026166	0.00210781
phase_3	3	0.0324848	0.0421219	0.0415322	0.0296828
phase_39	39	0.0296139	0.0462703	0.0489992	0.0240667
phase_40	40	0.0630137	0.0753723	0.082718	0.0607303
phase_78	78	0.216502	0.231871	0.254314	0.144051
phase_81	81	0.0709606	0.0910009	0.0962195	0.0699214
phase_156	156	1.30263	1.29614	1.31361	1.14426
phase_162	162	0.705221	0.738323	0.906911	0.585441
phase_240	240	0.853536	1.13628	1.08193	0.936212

Best-Cell Selection Ledger

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
ampl_0	0	SVR	0.000718839	LGBM	0.000988936	SVR (RMSE check: LGBM)
ampl_1	1	ERT	2.31427e-05	ERT	2.98514e-05	ERT
ampl_3	3	ERT	1.98433e-05	LGBM	2.68918e-05	ERT (RMSE check: LGBM)
ampl_39	39	LGBM	2.67456e-05	LGBM	4.1021e-05	LGBM
ampl_40	40	ERT	2.54591e-05	ERT	3.61431e-05	ERT
ampl_78	78	LGBM	3.68155e-05	LGBM	5.13891e-05	LGBM
ampl_81	81	LGBM	1.04092e-05	LGBM	1.53275e-05	LGBM
ampl_156	156	ERT	6.15641e-05	ERT	0.000234491	ERT
ampl_162	162	ERT	4.15775e-05	ERT	0.00011706	ERT
ampl_240	240	ERT	3.40349e-05	ERT	5.3833e-05	ERT
phase_1	1	LGBM	0.00158896	LGBM	0.00210781	LGBM
phase_3	3	LGBM	0.0227335	LGBM	0.0296828	LGBM
phase_39	39	LGBM	0.0175592	LGBM	0.0240667	LGBM
phase_40	40	ERT	0.0411526	LGBM	0.0607303	ERT (RMSE check: LGBM)
phase_78	78	LGBM	0.0758149	LGBM	0.144051	LGBM
phase_81	81	ERT	0.04529	ERT	0.0629514	ERT
phase_156	156	ERT	0.471632	ERT	0.96815	ERT

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
phase_162	162	DT	0.212884	LGBM	0.585441	DT (RMSE check: LGBM)
phase_240	240	DT	0.164686	DT	0.634265	DT

Backward Surface

Family columns: SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM .

Table 2 Equivalent - Amplitude MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.000919101	0.0146725	0.00168857	0.00228568	0.00114164	0.000803506
ampl_1	1	0.000177126	0.0120111	2.43462e-05	2.77814e-05	2.81504e-05	2.15653e-05
ampl_3	3	0.000131764	0.0129567	2.30055e-05	3.39017e-05	3.33253e-05	2.35706e-05
ampl_39	39	9.29819e-05	0.0178315	2.07116e-05	2.94403e-05	2.36151e-05	2.15845e-05
ampl_40	40	4.71358e-05	0.0155596	3.0577e-05	3.41385e-05	3.76091e-05	2.925e-05
ampl_78	78	0.000227798	0.0151595	6.39113e-05	9.35137e-05	7.76312e-05	4.66625e-05
ampl_81	81	4.24634e-05	0.0156248	1.24264e-05	1.69668e-05	1.4243e-05	9.66159e-06
ampl_156	156	0.00055626	0.00859988	0.000280313	0.00029556	0.000151484	9.32347e-05
ampl_162	162	0.000486365	0.00791997	0.000173519	0.000163426	6.25337e-05	4.61093e-05
ampl_240	240	0.000366943	0.0133052	0.00018752	0.000147515	8.21525e-05	9.38064e-05

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00138967	0.00204427	0.00271899	0.00112151
ampl_1	1	2.51784e-05	2.55221e-05	6.12132e-05	2.95045e-05
ampl_3	3	2.5368e-05	2.80158e-05	8.32641e-05	2.50467e-05
ampl_39	39	2.27117e-05	2.59831e-05	8.4512e-05	2.17983e-05
ampl_40	40	3.37425e-05	3.29784e-05	4.83798e-05	3.32849e-05
ampl_78	78	6.42575e-05	8.98032e-05	0.000145632	4.09575e-05
ampl_81	81	1.11522e-05	1.31939e-05	2.72327e-05	1.10453e-05
ampl_156	156	0.000240131	0.000367561	0.000388559	0.000348615
ampl_162	162	0.000127111	0.000202261	0.000254853	0.000140719
ampl_240	240	0.000195036	0.000221625	0.000255814	0.000121283

Table 3 Equivalent - Amplitude RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.00129485	0.0243713	0.00230014	0.00286206	0.00151495	0.00113874
ampl_1	1	0.000188894	0.0245844	3.33496e-05	3.74786e-05	3.97455e-05	2.91532e-05
ampl_3	3	0.00015476	0.0256323	3.59407e-05	4.65354e-05	5.78246e-05	3.98197e-05
ampl_39	39	0.000108285	0.0377068	3.02137e-05	3.90435e-05	3.15312e-05	3.32929e-05
ampl_40	40	5.84945e-05	0.0353067	4.463e-05	4.82786e-05	5.31668e-05	4.21787e-05
ampl_78	78	0.000263206	0.0387978	8.80374e-05	0.000121196	0.000120469	5.97958e-05
ampl_81	81	4.69126e-05	0.0358109	2.12304e-05	2.88106e-05	2.56393e-05	1.80578e-05
ampl_156	156	0.00213422	0.0224239	0.00121607	0.00122517	0.000531784	0.000284711
ampl_162	162	0.00146441	0.0154056	0.000595721	0.000595069	0.000217571	0.000158327
ampl_240	240	0.000991347	0.0280433	0.000636256	0.000281642	0.000194337	0.0003031

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00187843	0.00249674	0.0033211	0.00153081
ampl_1	1	3.42982e-05	3.66402e-05	9.82192e-05	3.92149e-05
ampl_3	3	3.74589e-05	4.39352e-05	0.000120464	3.48569e-05
ampl_39	39	3.19564e-05	3.76414e-05	0.00010196	2.98072e-05
ampl_40	40	4.8402e-05	4.58555e-05	5.86742e-05	4.81238e-05
ampl_78	78	8.69003e-05	0.000110976	0.000182553	5.63776e-05
ampl_81	81	1.97042e-05	2.16037e-05	3.73935e-05	1.71687e-05
ampl_156	156	0.00107075	0.00126722	0.00115926	0.0010952
ampl_162	162	0.000459473	0.000555421	0.000623493	0.000328885
ampl_240	240	0.000726963	0.000658981	0.0006855	0.000307316

Table 4 Equivalent - Phase MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00267855	0.0210761	0.00184404	0.00268244	0.00233752	0.00160364
phase_3	3	0.0257547	0.0648925	0.0235998	0.0294384	0.0275259	0.0236613
phase_39	39	0.851789	1.93686	0.347155	0.294936	0.309976	0.303497
phase_40	40	0.181417	0.210545	0.112521	0.126807	0.113752	0.0992531
phase_78	78	0.124318	0.141657	0.0718479	0.0880298	0.0669291	0.0674663
phase_81	81	0.140175	0.15748	0.0865766	0.140027	0.106607	0.0782247
phase_156	156	0.393344	0.570063	0.163951	0.149472	0.123047	0.12693
phase_162	162	0.311596	0.848346	0.06629	0.0869509	0.067229	0.05
phase_240	240	0.373308	0.639546	0.321272	0.281685	0.319162	0.274176

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00169253	0.0018153	0.00206844	0.00161948
phase_3	3	0.0250312	0.0294576	0.0353889	0.0221611
phase_39	39	0.313999	0.510234	0.549308	0.516687
phase_40	40	0.123007	0.139325	0.153344	0.12989
phase_78	78	0.0722265	0.0753545	0.0882764	0.0672852
phase_81	81	0.0945418	0.106594	0.113805	0.0998631
phase_156	156	0.168042	0.21939	0.237468	0.205108
phase_162	162	0.0704811	0.131111	0.132735	0.0709176
phase_240	240	0.241564	0.433959	0.380607	0.444441

Table 5 Equivalent - Phase RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00436848	0.0393695	0.0025878	0.00490168	0.00364965	0.00217166

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_3	3	0.0369946	0.0823274	0.0318584	0.0392277	0.0485335	0.0313362
phase_39	39	1.3701	2.29468	0.924331	1.23592	1.12725	0.914234
phase_40	40	0.331736	0.35206	0.171232	0.239004	0.222893	0.170365
phase_78	78	0.258024	0.197128	0.15697	0.152105	0.124797	0.173776
phase_81	81	0.177572	0.196575	0.113121	0.175858	0.145648	0.102423
phase_156	156	0.833131	0.935191	0.376142	0.359288	0.47852	0.389865
phase_162	162	0.591695	1.19694	0.109658	0.140077	0.117074	0.0930306
phase_240	240	0.971336	1.05413	0.689136	0.683308	0.848585	0.628844

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00244162	0.0026516	0.00285803	0.00212628
phase_3	3	0.032048	0.038754	0.0469444	0.0288473
phase_39	39	0.954905	1.05314	1.08897	1.07804
phase_40	40	0.1856	0.221865	0.241633	0.221686
phase_78	78	0.173605	0.156308	0.18818	0.13004
phase_81	81	0.123566	0.13785	0.143258	0.13273
phase_156	156	0.471731	0.462708	0.476764	0.419448
phase_162	162	0.12141	0.197097	0.198168	0.111126
phase_240	240	0.546893	0.868909	0.804209	0.782259

Best-Cell Selection Ledger

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
ampl_0	0	ERT	0.000803506	ERT	0.00113874	ERT
ampl_1	1	ERT	2.15653e-05	ERT	2.91532e-05	ERT
ampl_3	3	RF	2.30055e-05	LGBM	3.48569e-05	RF (RMSE check: LGBM)
ampl_39	39	RF	2.07116e-05	LGBM	2.98072e-05	RF (RMSE check: LGBM)
ampl_40	40	ERT	2.925e-05	ERT	4.21787e-05	ERT
ampl_78	78	LGBM	4.09575e-05	LGBM	5.63776e-05	LGBM
ampl_81	81	ERT	9.66159e-06	LGBM	1.71687e-05	ERT (RMSE check: LGBM)
ampl_156	156	ERT	9.32347e-05	ERT	0.000284711	ERT
ampl_162	162	ERT	4.61093e-05	ERT	0.000158327	ERT
ampl_240	240	ET	8.21525e-05	ET	0.000194337	ET
phase_1	1	ERT	0.00160364	LGBM	0.00212628	ERT (RMSE check: LGBM)
phase_3	3	LGBM	0.0221611	LGBM	0.0288473	LGBM
phase_39	39	DT	0.294936	ERT	0.914234	DT (RMSE check: ERT)
phase_40	40	ERT	0.0992531	ERT	0.170365	ERT
phase_78	78	ET	0.0669291	ET	0.124797	ET
phase_81	81	ERT	0.0782247	ERT	0.102423	ERT

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
phase_156	156	ET	0.123047	DT	0.359288	ET (RMSE check: DT)
phase_162	162	ERT	0.05	ERT	0.0930306	ERT
phase_240	240	GBM	0.241564	GBM	0.546893	GBM

Global Surface

Family columns: SVR, MLP, RF, DT, ET, ERT, GBM, HGBM, XGBM, LGBM .

Table 2 Equivalent - Amplitude MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.000952119	0.0123569	0.00138923	0.00249934	0.00121598	0.00085735
ampl_1	1	0.000172263	0.00937125	2.43546e-05	3.05963e-05	3.13903e-05	2.44521e-05
ampl_3	3	0.000169671	0.0118607	2.04426e-05	3.31179e-05	2.88679e-05	2.034e-05
ampl_39	39	0.000384593	0.00988981	2.55833e-05	4.69768e-05	3.61314e-05	2.1297e-05
ampl_40	40	0.000251705	0.0069115	2.79671e-05	3.50814e-05	3.67646e-05	2.46802e-05
ampl_78	78	0.000178166	0.0104736	5.57798e-05	0.000124393	6.6819e-05	4.01791e-05
ampl_81	81	0.000120663	0.00947878	1.25745e-05	1.85381e-05	1.57374e-05	1.05018e-05
ampl_156	156	0.000633804	0.00897898	0.000299189	0.000404669	0.000209541	0.00012836
ampl_162	162	0.000542837	0.00969656	0.000287426	0.000191809	9.75703e-05	6.66303e-05
ampl_240	240	0.000387305	0.0103057	0.000235949	0.000266841	0.000129711	8.8799e-05

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00145271	0.00123283	0.00164237	0.00143381
ampl_1	1	2.65155e-05	2.93592e-05	5.40641e-05	2.95027e-05
ampl_3	3	1.98532e-05	2.28777e-05	7.66e-05	2.49365e-05
ampl_39	39	2.25486e-05	2.41644e-05	8.98459e-05	2.55479e-05
ampl_40	40	2.86689e-05	3.18209e-05	6.58508e-05	3.0949e-05
ampl_78	78	4.85871e-05	4.18247e-05	0.000119272	4.47794e-05
ampl_81	81	1.16018e-05	1.28705e-05	4.78763e-05	1.27304e-05
ampl_156	156	0.000259117	0.000375414	0.000348531	0.000321621
ampl_162	162	0.000198718	0.000221641	0.000235306	0.000171812
ampl_240	240	0.000137712	0.000161493	0.000211423	0.000157861

Table 3 Equivalent - Amplitude RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
ampl_0	0	0.00230658	0.0170706	0.00252125	0.00347712	0.00284096	0.0022606
ampl_1	1	0.000187927	0.018	3.48408e-05	4.43474e-05	4.57402e-05	3.46893e-05
ampl_3	3	0.000205732	0.0213407	2.83103e-05	4.43059e-05	4.3791e-05	3.1055e-05
ampl_39	39	0.000412607	0.019261	3.57047e-05	6.33515e-05	5.43275e-05	3.01315e-05
ampl_40	40	0.000264557	0.0129018	4.37554e-05	5.49025e-05	5.38109e-05	3.68428e-05
ampl_78	78	0.000213004	0.0182424	7.77069e-05	0.000160116	9.31048e-05	5.78135e-05
ampl_81	81	0.000145078	0.0172119	1.9526e-05	2.8769e-05	2.26808e-05	1.57609e-05
ampl_156	156	0.00218105	0.0157516	0.00107638	0.00159324	0.000801799	0.000448102
ampl_162	162	0.00177879	0.0170749	0.000831718	0.00050668	0.000289608	0.000229119
ampl_240	240	0.000891887	0.0190486	0.000598747	0.000842952	0.000381304	0.000205047

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
ampl_0	0	0.00280631	0.00257234	0.00277615	0.00265315
ampl_1	1	3.71924e-05	4.0223e-05	7.32885e-05	3.90002e-05
ampl_3	3	2.76399e-05	3.10717e-05	9.83373e-05	3.38989e-05
ampl_39	39	3.27234e-05	3.27633e-05	0.000118319	3.4056e-05
ampl_40	40	4.35675e-05	4.65812e-05	8.61133e-05	4.30988e-05
ampl_78	78	6.83023e-05	5.64841e-05	0.000161752	5.74562e-05
ampl_81	81	1.71217e-05	1.8916e-05	6.3698e-05	1.87623e-05
ampl_156	156	0.000963085	0.00163122	0.00132806	0.00103849
ampl_162	162	0.000600679	0.000771984	0.000639512	0.000399953
ampl_240	240	0.000343282	0.000305001	0.000457843	0.000331414

Table 4 Equivalent - Phase MAE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00266976	0.00950975	0.0022415	0.00261195	0.0028707	0.0020758
phase_3	3	0.0335132	0.0709203	0.0253337	0.0385292	0.0293429	0.0220847
phase_39	39	0.316534	0.514087	0.196247	0.217507	0.235392	0.120094
phase_40	40	0.108309	0.17468	0.0683166	0.0989789	0.0814536	0.0618699
phase_78	78	0.13399	0.105781	0.0566897	0.126693	0.0688349	0.047961
phase_81	81	0.0904747	0.129788	0.0674708	0.113506	0.0776575	0.061698
phase_156	156	0.825491	1.05057	0.48731	0.619049	0.346379	0.34956
phase_162	162	0.322757	0.722826	0.126892	0.144198	0.122334	0.0842359
phase_240	240	0.645422	0.705658	0.443941	0.531634	0.403805	0.32347

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00218867	0.00229796	0.00232539	0.00220298
phase_3	3	0.0248632	0.0269955	0.0240916	0.0270174
phase_39	39	0.18782	0.215929	0.213635	0.220116
phase_40	40	0.0662234	0.0826545	0.0764415	0.0817864
phase_78	78	0.0522307	0.0650381	0.0687188	0.0816808
phase_81	81	0.0629744	0.0762587	0.0760646	0.0769727
phase_156	156	0.431761	0.520743	0.531617	0.555527
phase_162	162	0.105331	0.171532	0.137712	0.170437
phase_240	240	0.413962	0.524518	0.462971	0.552085

Table 5 Equivalent - Phase RMSE

Panel 1 of 2: SVR, MLP, RF, DT, ET, ERT .

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_1	1	0.00407776	0.0155811	0.00337943	0.00429518	0.00492447	0.00317864

Target	Harmonic	SVR	MLP	RF	DT	ET	ERT
phase_3	3	0.0514158	0.090849	0.0399168	0.0569582	0.0455011	0.0362186
phase_39	39	0.741814	0.922865	0.641498	0.972638	1.00277	0.535965
phase_40	40	0.222343	0.273519	0.118132	0.149025	0.129478	0.110887
phase_78	78	0.27125	0.156399	0.122674	0.260166	0.142941	0.111965
phase_81	81	0.125372	0.169488	0.0950763	0.159991	0.115755	0.0905346
phase_156	156	1.35775	1.46808	1.00215	1.38891	0.927969	0.794264
phase_162	162	0.575087	1.07561	0.268824	0.448358	0.502456	0.209722
phase_240	240	1.42493	1.24579	1.10474	1.22484	1.21213	0.927856

Panel 2 of 2: GBM, HGBM, XGBM, LGBM .

Target	Harmonic	GBM	HGBM	XGBM	LGBM
phase_1	1	0.00326404	0.00332204	0.00345103	0.00316819
phase_3	3	0.0397177	0.0445096	0.0392022	0.0434704
phase_39	39	0.690014	0.698951	0.614245	0.681693
phase_40	40	0.114052	0.132295	0.12726	0.130387
phase_78	78	0.101854	0.118781	0.137359	0.161298
phase_81	81	0.0939085	0.103452	0.112958	0.101726
phase_156	156	0.953602	1.06293	1.06696	1.04266
phase_162	162	0.236531	0.344645	0.273063	0.371405
phase_240	240	1.12535	1.08091	1.09298	1.15704

Best-Cell Selection Ledger

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
ampl_0	0	ERT	0.00085735	ERT	0.0022606	ERT
ampl_1	1	RF	2.43546e-05	ERT	3.46893e-05	RF (RMSE check: ERT)
ampl_3	3	GBM	1.98532e-05	GBM	2.76399e-05	GBM
ampl_39	39	ERT	2.1297e-05	ERT	3.01315e-05	ERT
ampl_40	40	ERT	2.46802e-05	ERT	3.68428e-05	ERT
ampl_78	78	ERT	4.01791e-05	HGBM	5.64841e-05	ERT (RMSE check: HGBM)
ampl_81	81	ERT	1.05018e-05	ERT	1.57609e-05	ERT
ampl_156	156	ERT	0.00012836	ERT	0.000448102	ERT
ampl_162	162	ERT	6.66303e-05	ERT	0.000229119	ERT
ampl_240	240	ERT	8.8799e-05	ERT	0.000205047	ERT
phase_1	1	ERT	0.0020758	LGBM	0.00316819	ERT (RMSE check: LGBM)
phase_3	3	ERT	0.0220847	ERT	0.0362186	ERT
phase_39	39	ERT	0.120094	ERT	0.535965	ERT
phase_40	40	ERT	0.0618699	ERT	0.110887	ERT
phase_78	78	ERT	0.047961	GBM	0.101854	ERT (RMSE check: GBM)
phase_81	81	ERT	0.061698	ERT	0.0905346	ERT

Target	Harmonic	MAE fam	MAE	RMSE fam	RMSE	Candidate
phase_156	156	ET	0.346379	ERT	0.794264	ET (RMSE check: ERT)
phase_162	162	ERT	0.0842359	ERT	0.209722	ERT
phase_240	240	ERT	0.32347	ERT	0.927856	ERT